

Mohamad ElMahdi Houmani

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<https://github.com/mahdihoumanii> Date of Birth: 19 May 2001

PROFILE

Highly analytical and solution-driven MSc Physics graduate with a strong foundation in mathematical modeling, computational physics, and financial mathematics. Demonstrates robust skills in quantitative analysis, problem-solving, and collaboration. Adept at building efficient models and delivering data-driven insights.

EDUCATION

- **RWTH Aachen University**, Germany
Master's in Data Science 2026
- **RWTH Aachen University**, Germany
Master's in Physics (Quantum Field Theory and Gauge Theories) 2022 – 2024
- **Lebanese University**, Lebanon
Bachelor's in General Physics 2018 – 2021

SELECTED PROJECTS

- Volatility Modeling and Option Pricing** 2025
Developed and evaluated rolling, EWMA, and GARCH(1,1) volatility models using walk-forward testing and applied forecasts to European option pricing via Black–Scholes and Monte Carlo simulation. GitHub
- Heston Stochastic Volatility: Pricing & Calibration** 2025
Implemented Heston pricing via characteristic functions, implied volatility inversion, and calibrated parameters to option surfaces using nonlinear optimization with no-arbitrage and robustness diagnostics. GitHub
- Portfolio Management and Risk Allocation** 2025
Built an end-to-end walk-forward portfolio backtesting framework implementing risk-based allocation strategies with transaction costs and comprehensive performance and risk diagnostics. GitHub

PROFESSIONAL EXPERIENCE

Student Research Assistant — RWTH Aachen University, Germany

- **Theoretical Particle Physics Department (2024)**: Computed massive five-point amplitudes in QCD using finite-field methods; expanded in small transverse momentum and compared results with MadGraph..
- **Mechanical Engineering and Combustion Technology (2022–2024)**: Researched turbulent combustion using MATLAB and experimental tools; presented findings effectively.
- **Particle Physics and Cosmology Group (2022–2023)**: Simulated gravitational effects on mirrors for the Einstein Telescope using Python.

KEY SKILLS

- **Mathematical & Financial Modeling**: Stochastic processes, regression analysis, time series, portfolio theory.
- **Programming**: Python (NumPy, Pandas, SciPy), C++, MATLAB, Mathematica.
- **Technical Tools**: FORM, TAPIR, MadGraph, Monte Carlo simulation, symbolic computation.
- **Soft Skills**: Agile development, cross-disciplinary teamwork, technical communication, academic research.

VOLUNTEER EXPERIENCE

- **RWTH Aachen — International Student Support** 2022
Supported international students in navigating university systems and integrating socially.

LANGUAGES

English: Fluent **French**: Fluent **Arabic**: Native **German**: B2